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B.Tech. Degree I & II Semester Regular/Supplementary Examination September 2022

CE/EE/ME/SE 19-200-0205A / CS/EC/IT 19-200-0105B BASIC MECHANICAL ENGINEERING
(2019 Scheme)

Time: 3 Hours

Maximum Marks: 60

Course Outcomes

On successful completion of the course, the students will be able to:

- CO1: Understand basics of thermodynamics and working of steam turbines.
 CO2: Understand basics internal combustion engines, refrigeration and air conditioning.
 CO3: Gain knowledge on the working of hydraulic turbines and centrifugal pumps.
 CO4: Identify manufacturing methods encountered in engineering practice and understand mechanism of power transmission.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze,
 L5 – Evaluate, L6 – Create
 PO – Programme Outcome

PART A (Answer ALL questions)

	(8 × 3 = 24)	Marks	BL	CO	PO
I. (a) Explain Zeroth law of thermodynamics. Explain its significance.	3	3	L3	1	1,2
(b) Differentiate between Heat and Work and state their sign conventions.	3	3	L1	1	1,2
(c) Explain Carburetor with the help of a neat figure.	3	3	L2	2	1
(d) Explain the four physical process involved in Air Conditioning.	3	3	L2	2	1,2
(e) Define the specific speed of a turbine. What is the significance of the specific speed?	3	3	L1	3	1,2
(f) What are the advantages and disadvantages of the following power plants? (i) Thermal (ii) Nuclear	3	3	L1	3	1,2
(g) List down the typical applications of the following processes: (i) Casting (ii) Forging (iii) Rolling	3	3	L2	4	1
(h) Explain the difference between annealing and normalising.	3	3	L2	4	1,2

PART B

(4 × 12 = 48)

II. (a) 1 Kg of air at a pressure of 6 bar and a temperature of 40°C expands to a pressure of 1 bar according to the law $PV^{1.3}$ is constant. Determine (i) Change in internal energy (ii) Work Done (iii) Heat supplied.	6	6	L4	1	2,3
(b) State and prove the equivalence of second law of thermodynamics.	6	6	L4	1	3,4
OR					
III. (a) Explain any one type of steam generator with the help of a neat sketch.	6	6	L1	1	3,4
(b) Explain with neat figures compounding of steam turbines.	6	6	L1	1	3,4

(P.T.O.)

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		Marks	BL	CO	PO
IV.	(a) An ideal diesel cycle operates on 1 kg of air with an initial pressure of 1 bar and a temperature of 35°C. The pressure at the end of compression is 33 bar and cut off is 6% of the stroke. Determine (i) Compression ratio (ii) Heat supplied (iii) Heat rejected.	6	L5	2	3,4
	(b) Explain with neat figure the cooling systems used in IC engines.	6	L2	2	3,4
OR					
V.	(a) Explain simple vapour compression refrigeration system with the help of a neat figure.	6	L4	2	3,4
	(b) Explain Summer Air conditioning and Winter Air conditioning with the help of a neat figure.	6	L4	2	3,4
VI.	(a) Briefly explain the constructional details and working of a hydraulic turbine which is suitable for low head applications.	6	L2	3	3,4
	(b) List the main components of a centrifugal pump and explain them briefly.	6	L2	3	3,4
OR					
VII.	(a) With the help of a simple layout diagram, explain the working of a hydro electric plant. What are the functions of the major components in the plant?	6	L2	3	3,4
	(b) Write short notes on (i) Wind mills (ii) Solar energy.	6	L2	3	3,4
VIII.	(a) Explain the process of Arc welding with the help of a sketch.	6	L2	4	4,5
	(b) Explain the extrusion process. Compare direct and indirect extrusion process.	6	L2	4	3,4
OR					
IX.	(a) Explain the relative merits and demerits of Gear, Rope and Chain Drive for transmission of power.	6	L2	4	4,5
	(b) Two parallel shafts six meters apart are provided with 60 cm and 80 cm diameter pulleys and are connected by means of a cross belt. The direction of rotation of follower pulley is to be reversed by changing over to an open belt drive. How much length of belt has to be reduced?	6	L2	4	3,4

Bloom's Taxonomy Levels

L1 - 17.5%, L2 - 50%, L3 - 2.5%, L4 - 20%, L5 - 10%.
